TEST PLAN

ETL Full Extraction — CodingSynonym Exclusion Validation

Module	ETL — Full Extraction Pipeline
Feature Under Test	Exclusion of CodingSynonym records from ETL output
Execution Type	CLI ETL via Jenkins Pipeline
Extraction Modes Covered	Full Extraction (Lookup & Dictionary auto-triggered)
Target Tables Validated	
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Document Version	1.0

1. Scope & Objective

This test plan validates that the ETL Full Extraction pipeline, when executed via CLI through the Jenkins pipeline, correctly excludes CodingSynonym data from the extracted output. Specifically, it confirms two acceptance criteria:

(1) The *CodingSynonym_tbl* must contain zero records after a Full ETL run for the target study, and

(2) The *AuditValueChange_tbl* must not contain any records where *TargetObjectType = 'CodingSynonym'*.

This plan also covers regression checks to ensure that valid coded variable data and non-CodingSynonym audit records are not inadvertently suppressed.

2. Test Pre-Conditions & Setup

All five pre-conditions below must be satisfied and verified before executing any test case. Skipping these steps will produce invalid results.

#	Pre-condition Step	Details / Expected State
PC-01	Study with Coded Form Variables	1. Select a study that contains at least one form with codeable variables. 2. Code a minimum of 2–3 variables (e.g., MedDRA-coded AE term, WHO Drug-coded concomitant medication).
PC-02	Coding Synonyms Created	1. While coding each variable, add at least 1–2 Coding Synonyms per coded term. 2. Verify synonyms are persisted and visible in the Coding Synonym list within the application before proceeding.
PC-03	ETLRunLog Cleared for the Study	1. Execute the following before running ETL: 2. DELETE FROM ETLRunLog WHERE StudyID = '<TargetStudyID>'; 3. Confirm row count = 0 for that StudyID post-deletion. 4. This ensures no residual run entries interfere with the new execution.
PC-04	Jenkins Pipeline Configured for Study	1. The Jenkins CLI ETL job must be configured with the correct StudyOID / StudyID parameter. 2. Confirm the pipeline includes Lookup & Dictionary extraction steps (auto-invoked on Full Extraction runs).
PC-05	Baseline DB Snapshot (Optional but Recommended)	1. Record pre-ETL counts for reference: 2. SELECT COUNT(*) FROM CodingSynonym_tbl; 3. SELECT COUNT(*) FROM AuditValueChange_tbl WHERE TargetObjectType = 'CodingSynonym';

3. Test Case Summary

The following eight test cases cover the two acceptance criteria, their inverse regression checks, automatic sub-extraction verification, and a repeat-run stability test.

TC ID	Test Objective	Type	Priority	Linked AC	Auto / Manual
TC-ETL-001	CodingSynonym_tbl must be empty after Full ETL run	Negative	Critical	AC-1	Manual
TC-ETL-002	AuditValueChange_tbl must have zero records where TargetObjectType = 'CodingSynonym'	Negative	Critical	AC-2	Manual
TC-ETL-003	Jenkins Full ETL triggers Lookup extraction automatically	Functional	High	Setup	Manual
TC-ETL-004	Jenkins Full ETL triggers Dictionary extraction automatically	Functional	High	Setup	Manual
TC-ETL-005	Coded variable data (non-synonym) is correctly extracted	Regression	High	AC-1 (inverse)	Manual
TC-ETL-006	ETLRunLog is populated correctly after Full ETL execution	Functional	Medium	Setup	Manual
TC-ETL-007	AuditValueChange_tbl still contains records for other TargetObjectTypes post-ETL	Regression	High	AC-2 (inverse)	Manual
TC-ETL-008	Re-run ETL after adding additional Coding Synonyms — table remains unpopulated	Regression	Medium	AC-1, AC-2	Manual

TC-ETL-001 CodingSynonym_tbl — Verify Zero Records After Full ETL

Precondition	PC-01 through PC-04 completed. At least 1 Coding Synonym created and confirmed in-app.		
Test Steps	Step 1: Clear ETLRunLog for StudyID (as per PC-03). Step 2: Trigger Jenkins pipeline: run CLI ETL Full Extraction for the target study. Step 3: Wait for the pipeline to complete successfully (all stages GREEN in Jenkins). Step 4: Connect to the ETL output / staging database. Step 5: Execute the validation SQL against CodingSynonym_tbl.	**Validation SQL**	SELECT COUNT(*) FROM CodingSynonym_tbl WHERE StudyID = '<TargetStudyID>';
Expected Result	COUNT(*) = 0 for the target study. No records from CodingSynonym_tbl should appear in the ETL output dataset. The table should have no rows populated regardless of how many synonyms were created in the application.		
Pass / Fail	[ ] Pass [ ] Fail [ ] Blocked		

TC-ETL-002 AuditValueChange_tbl — No Records with TargetObjectType = 'CodingSynonym'

Precondition	PC-01 through PC-04 completed. At least 2 Coding Synonyms created, each triggering audit entries in the application.		
Test Steps	Step 1: After Full ETL Jenkins pipeline completes, connect to the output database. Step 2: Query AuditValueChange_tbl with the filter TargetObjectType = 'CodingSynonym'. Step 3: Scope the query to the target study to avoid cross-study false positives. Step 4: Also check without StudyID filter to confirm global absence.	**Validation SQL**	SELECT COUNT(*) FROM AuditValueChange_tbl WHERE TargetObjectType = 'CodingSynonym' AND StudyID = '<TargetStudyID>';
Expected Result	COUNT(*) = 0 — absolutely zero audit records must exist where TargetObjectType = 'CodingSynonym'. This confirms that CodingSynonym-related audit events are intentionally suppressed during extraction. Other TargetObjectType values (e.g., 'DataPoint', 'Subject') must remain unaffected.		
Pass / Fail	[ ] Pass [ ] Fail [ ] Blocked		

TC-ETL-003 Lookup Extraction Auto-Triggered via Jenkins Full ETL

Precondition	Jenkins pipeline configured. Full Extraction mode selected (not incremental).

Test Steps	Step 1: Trigger CLI ETL Full Extraction via Jenkins for the study. Step 2: Monitor the Jenkins console output for the Lookup extraction stage. Step 3: Confirm the Lookup extraction job log entry appears in ETLRunLog. Step 4: Verify Lookup tables are populated in the output database.	**Validation SQL**	SELECT ExtractionType, Status FROM ETLRunLog WHERE StudyID= '<TargetStudyID>' AND ExtractionType = 'Lookup';
Expected Result	ETLRunLog must contain a row with ExtractionType = 'Lookup' and Status = 'Completed' (or equivalent success status). Lookup tables in the output schema must be populated with at least one record. No manual trigger of Lookup extraction should be required.		
Pass / Fail	[ ] Pass [ ] Fail [ ] Blocked		

TC-ETL-004 Dictionary Extraction Auto-Triggered via Jenkins Full ETL			
Precondition	Study has at least one coded variable using a coding dictionary (e.g., MedDRA or WHO-DD).		
Test Steps	Step 1: Trigger the same Jenkins Full ETL pipeline used in TC-ETL-003. Step 2: Monitor Jenkins logs for the Dictionary extraction stage execution. Step 3: Confirm ETLRunLog contains a Dictionary extraction entry. Step 4: Verify Dictionary-related tables are populated in the output database.	**Validation SQL**	SELECT ExtractionType, Status FROM ETLRunLog WHERE StudyID= '<TargetStudyID>' AND ExtractionType = 'Dictionary';
Expected Result	ETLRunLog must contain a row with ExtractionType = 'Dictionary' and Status = 'Completed'. Dictionary tables in output schema must reflect the dictionary version assigned to the study. Lookup and Dictionary extractions must both appear in the same run — triggered by the single Full ETL invocation.		
Pass / Fail	[ ] Pass [ ] Fail [ ] Blocked		

TC-ETL-005 Coded Variable Data (Non-Synonym) Is Correctly Extracted			
Precondition	PC-01 and PC-02 completed. Coded variables have both coding results and synonyms.		
Test Steps	Step 1: After Full ETL completes, query the coded data output table for the study. Step 2: Confirm rows exist for the coded variables (AE term, medication, etc.). Step 3: Cross-check that the coded value (e.g., MedDRA PT code, WHO Drug code) is present. Step 4: Confirm the CodingSynonym is NOT embedded or joined into these rows.	**Validation SQL**	SELECT COUNT(*) FROM CodedData_tbl WHERE StudyID = '<TargetStudyID>' AND CodedValue IS NOT NULL;
Expected Result	COUNT(*) > 0 — coded variable data must be extracted and present in the output. The exclusion of CodingSynonym_tbl must not inadvertently suppress coded variable records themselves. This acts as a regression guard to confirm selective exclusion (synonyms only) is working.		
Pass / Fail	[ ] Pass [ ] Fail [ ] Blocked		

TC-ETL-006 ETLRunLog Populated Correctly After Full ETL Execution			
Precondition	ETLRunLog cleared (PC-03). Full ETL run completed via Jenkins.		
Test Steps	Step 1: After Jenkins pipeline completion, query ETLRunLog for the study. Step 2: Confirm at minimum three run entries: Full, Lookup, and Dictionary. Step 3: Verify StartTime, EndTime, and Status fields are all populated. Step 4: Confirm no failed or stuck status values exist for this run.	**Validation SQL**	SELECT ExtractionType, StartTime, EndTime, Status FROM ETLRunLog WHERE StudyID = '<TargetStudyID>' ORDER BY StartTime;
Expected Result	ETLRunLog must contain rows for: ExtractionType IN ('Full', 'Lookup', 'Dictionary'). All rows must show Status = 'Completed' (or the application's success status value). StartTime and EndTime must both be non-null and EndTime > StartTime. No rows with Status = 'Failed' or Status = 'Running' should remain post-completion.		

Pass / Fail	[ ] Pass [ ] Fail [ ] Blocked

TC-ETL-007 AuditValueChange_tbl Retains All Non-CodingSynonym Records			
Precondition	Study must have other audit-generating events: data entry changes, subject status changes, etc.		
Test Steps	Step 1: After Full ETL, query AuditValueChange_tbl for the study. Step 2: Filter for TargetObjectType != 'CodingSynonym' and confirm rows exist. Step 3: Compare counts against pre-ETL baseline recorded in PC-05. Step 4: Validate that TargetObjectType values like 'DataPoint', 'Subject', 'Form' are present.	**Validation SQL**	SELECT TargetObjectType, COUNT(*) AS cnt FROM AuditValueChange_tbl WHERE StudyID = '<TargetStudyID>' GROUP BY TargetObjectType;
Expected Result	Non-CodingSynonym audit records must be present and match expected baseline counts. The ETL must not over-suppress audit records — only CodingSynonym entries are excluded. Result set must show at least one non-CodingSynonym TargetObjectType with COUNT > 0.		
Pass / Fail	[ ] Pass [ ] Fail [ ] Blocked		

TC-ETL-008 Re-run ETL After Adding More Synonyms — CodingSynonym_tbl Remains Empty			
Precondition	TC-ETL-001 passed. Add 2–3 additional Coding Synonyms to the same or different coded variables.		
Test Steps	Step 1: After TC-ETL-001 passes, return to the application and add additional Coding Synonyms. Step 2: Clear ETLRunLog again for the study. Step 3: Re-run CLI ETL Full Extraction via Jenkins for the same study. Step 4: Re-execute TC-ETL-001 and TC-ETL-002 validation SQLs.	**Validation SQL**	SELECT COUNT(*) FROM CodingSynonym_tbl WHERE StudyID = '<TargetStudyID>'; -- AND -- SELECT COUNT(*) FROM AuditValueChange_tbl WHERE TargetObjectType ='CodingSynonym';
Expected Result	CodingSynonym_tbl COUNT(*) = 0 even after incremental addition of synonyms. AuditValueChange_tbl CodingSynonym count = 0 on repeated run. This confirms the exclusion is persistent and not a one-time artifact of the initial empty state.		
Pass / Fail	[ ] Pass [ ] Fail [ ] Blocked		

4. Edge Case & Boundary Scenarios

The following scenarios must be tested in addition to the primary test cases to ensure robustness of the fix across edge conditions.

Scenario	Validation Action	Expected Outcome
Coding Synonym added immediately before ETL trigger (no delay)	Trigger ETL within 60 seconds of creating the synonym; check CodingSynonym_tbl post-run.	COUNT(*) = 0. No race condition should allow synonym to slip through.
Multiple Coding Synonyms exist across multiple forms and multiple studies	Run ETL for a single study; check both that study and others in CodingSynonym_tbl.	Only the target study is included in the run. No cross-contamination from other studies.
ETLRunLog not cleared before re-run (stale log entries present)	Run ETL without clearing log; check if the pipeline completes normally and does not re-extract CodingSynonym_tbl.	Pipeline must still complete. CodingSynonym_tbl = 0. Stale log must not cause synonym data leakage.
AuditValueChange_tbl queried without StudyID filter (global check)	Run: SELECT COUNT(*) FROM AuditValueChange_tbl WHERE	COUNT(*) = 0 globally. No CodingSynonym audit entries should exist in the entire table across all studies.

	TargetObjectType = 'CodingSynonym'; (no study filter)	

5. Defect Logging Guidance

If any test case fails, capture the following evidence before raising a defect ticket. All critical failures must block sign-off on the technical task.

Failure Scenario	Defect Severity	Evidence to Capture
CodingSynonym_tbl returns COUNT(*) > 0 after Full ETL	**Critical**	SQL query result screenshot Jenkins console output (full log) ETLRunLog rows for the study run DB record samples from CodingSynonym_tbl
AuditValueChange_tbl has records where TargetObjectType = 'CodingSynonym'	**Critical**	SQL query result with COUNT and sample rows TargetObjectType distinct values from same table Jenkins run ID and timestamp
Lookup or Dictionary extraction did NOT auto-trigger during Full ETL	**High**	Jenkins pipeline stage view screenshot ETLRunLog — missing ExtractionType row Full Jenkins console log
Coded variable data absent after ETL (synonym exclusion over-suppressed)	**High**	COUNT from coded data output table Pre-ETL baseline vs post-ETL comparison Form and variable details where coding was done